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Dear Editors of *Acta Informatica*,

I am pleased to submit the manuscript entitled "Real-Time $p{:}q$ Deterministic Pushdown Automata: Generalization, Normalization, and Exact Control Width – A WRITE–SEARCH–CANCEL Theory for Weighted Binary Balance Languages" for consideration as a Research Article in *Acta Informatica*.

The paper starts from a completion-aware deterministic pushdown automaton developed for the binary counting language $L_{2,1} = \{w : N_1(w) = 2N_0(w)\}$. Rather than treating that machine as an isolated construction, the manuscript extracts its completion semantics and develops a uniform real-time architecture for every coprime positive ratio $p{:}q$ in the family $L_{p,q} = \{w : qN_1(w) = pN_0(w)\}$.

The general construction stores literal shortest-completion demand on a binary work stack and transfers one-cell LIFO mismatch information into finite control as signed search debt. A primitive zero-weight kernel $K = (q, p)$ determines the normalization arithmetic. The manuscript proves a residual invariant, exact projection to the unique minimum completion demand, a canonical-segment invariant yielding two-cell qualitative visibility, exact WRITE normalization, and zero-residual correctness. Within an explicitly defined canonical deferred-SEARCH realization model, it also proves a matching lower bound giving exact control width $2\lfloor(p + q - 1)/2\rfloor + 1$.

A further structural result closes the construction loop. Specializing the general machine back to $p{:}q = 2{:}1$ recovers the original $C/L/R$ control architecture, the state potentials, cancellation/search roles, and characteristic WRITE arithmetic. The two remaining transition-family differences are shown to be a normalization-schedule choice: the historical machine discharges those debts eagerly, while the centered specialization keeps them lazily in finite control, with identical shortest-demand semantics.

The scope of the lower bound is stated explicitly; the manuscript does not claim unrestricted global state optimality for all DPDAs recognizing $L_{p,q}$. Likewise, established work on deterministic one-counter automata, pushdown descriptional complexity, and shortest-path phenomena is treated as background rather than claimed as new.

I wish to disclose the closely related companion manuscript "Completion Frontiers and Quotient Geometry in Deterministic Pushdown Automata for Rational Binary Counting." Both works originate from the same completion-aware 2:1 design and share the residual/completion viewpoint. The present submission has distinct primary results: the centered real-time literal-demand construction for arbitrary coprime $p{:}q$, its canonical-segment/depth-two invariant, the exact deferred-SEARCH control-width theorem, and the specialization/reconstruction result with eager–lazy normalization equivalence.

The manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. I am the sole author and take responsibility for the submitted work.

Thank you for considering this manuscript for publication in *Acta Informatica*.

Sincerely,

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